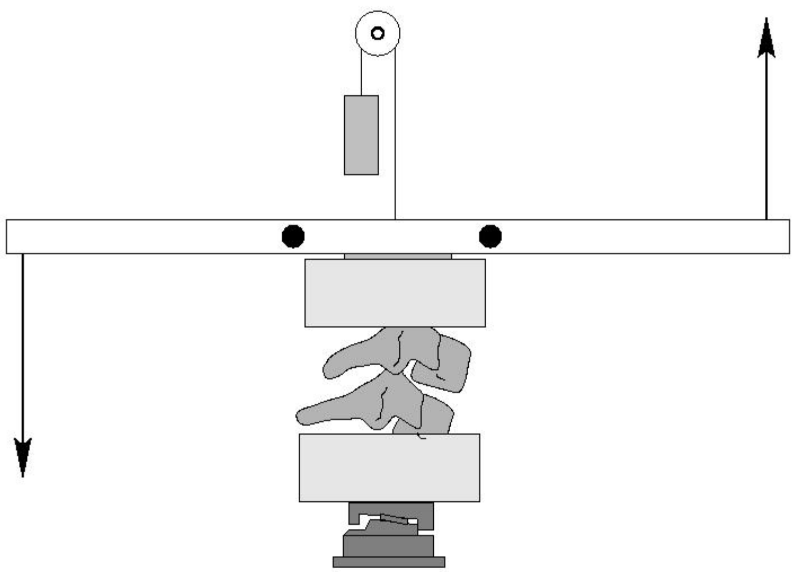
Bending Responses of The Human Cervical Spine

Failure Test Report

Test Reference: B26MO2FAIF

Test Date: 1/22/2003 Submission Date: 11/1/2003



Duke University, Department of Biomedical Engineering

NHTSA Cooperative Agreement No. *DTNH22-94-Y-07133*

Investigators:

Roger Nightingale, Ph.D.
Barry S. Myers, M.D., Ph.D.

Test Performers:

Laura Tran
Danielle Ottaviano

Study Objective:

The objective of this study is to provide data on the responses of the human cervical spine to pure bending. The experimental protocol consists of three tests: two flexibility tests and a failure test. The flexibility tests are designed to quantify the static response of motion segments to flexion and extension moments. The failure test is designed to quantify the bending tolerance of motion segments. See the Methods Section at the end of this report for more details on the experimental protocol.

Specimen ID: B26M

Test Reference: B26MO2FAIF

Test Objective: Flexion Failure

Spine Segment: Occiput - C2

Specimen Description

Sex:	*Male*	Age:	*74 years*
Weight:	*N/A (N)*	Height:	*N/A (cm)*

Comments: Deep harvest cut at C6-C7. Major osteophyte in C4-C5 disc space. Only OC2 is testable.

Injury Summary

Noncatastrophic injury. Both posterior atlanto-axial membranes were torn. Capsules at right occipital condyle were torn along with small broken bone spur (probably from joint ossification).

(Y-axis) Moment at Failure: ***33.10** N-m*
Angle at Failure: ***61.80** °*

Video Data:

All the digital images are included on the CD which accompanies this report. They are in *tif* format and can be found in the subdirectory, *HSVIDEO*. These represent sequential images, with the frame 0000 corresponding to time 0 of the instrument data. All the angles are computed relative to the specimen Reference Image. The Reference Image is included with the flexibility test report for this specimen.

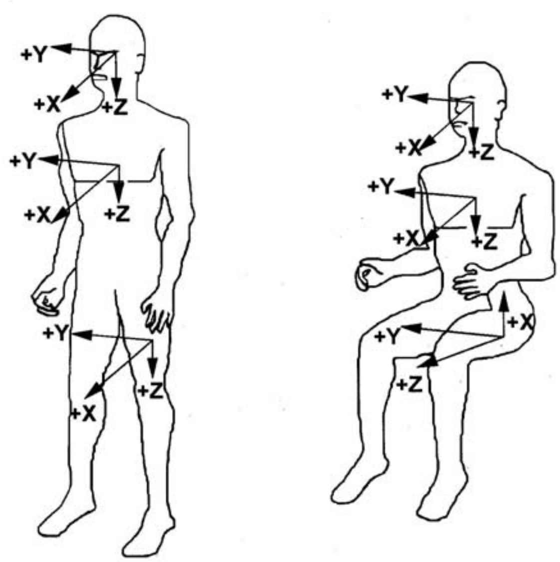
Camera Frame Rate: 125 (frames per second)

Test Comments: No comments.

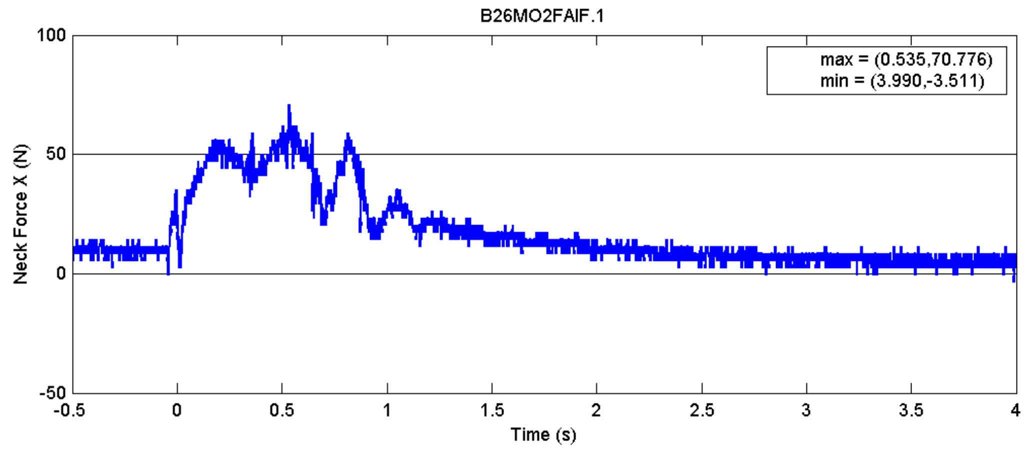
Appendix A: Coordinate system and sign conventions

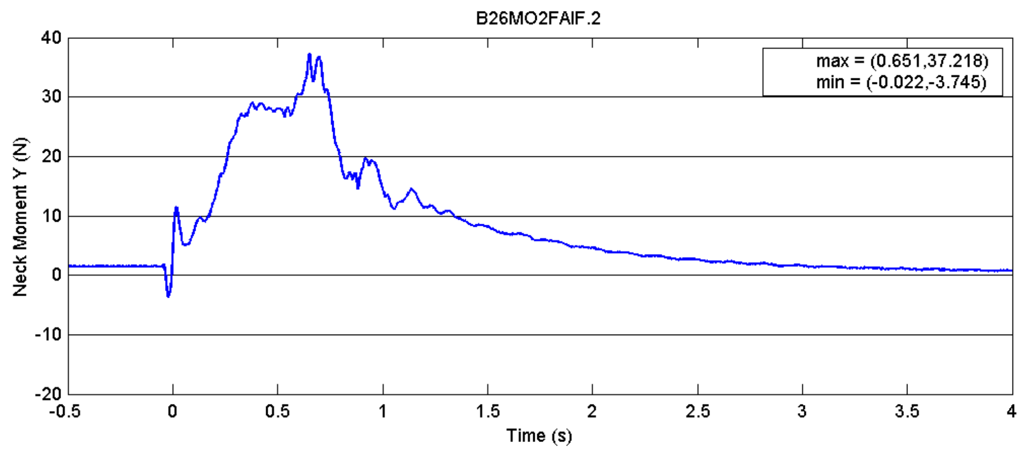
Sign Conventions: Compliant with SAE J211/1 March 1995

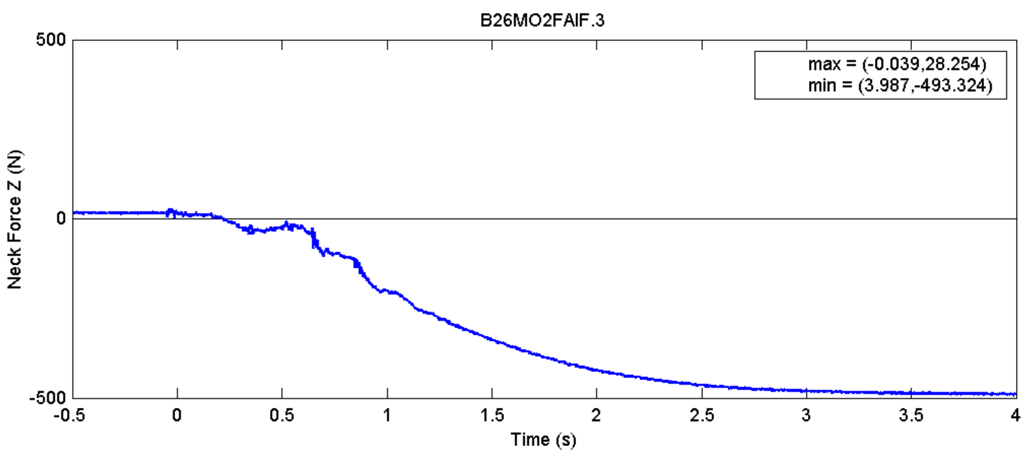
Load Cell	Measure	Manipulation	Polarity
Denton 1110AF-5K Upper Neck Load Cell	Fx	Head Rearward	+
	My	Chin to Sternum	+
	Fz	Head Upward	+



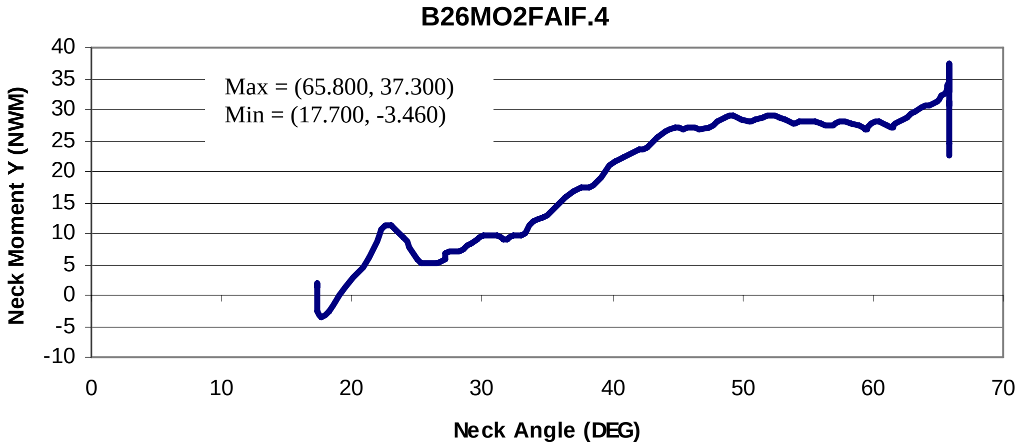
Appendix B: Instrument Data







Appendix C: Computed Data



Appendix D: Instrumentation

Instrumentation

Channel	Transducer	Measure	Manipulation	Axis
1	Denton 1110AF-5K	Fx (N)	Head Rearward	+x
2	Upper Neck Load Cell	My (N-m)	Chin to Sternum	+y
3		Fz (N)	Head Upward	+z

Data Integrity

Channel	Clip	Spike	Fail
1	☐	☐	☐
2	☐	☐	☐
3	☐	☐	☐

Data Acquisition:

Sampling Frequency: ***1000** (hz)*   Number of Samples: ***4500***

Computed Data:

Channel	Computation	Measure	Manipulation	Axis
4	Flexibility	Angle (°)	Chin to Sternum	+y

Methods

Unembalmed human heads with intact spines were obtained shortly after death. All specimen handling was performed in compliance with both NHTSA 700-4 guidelines and CDC guidelines. The specimens were sealed in plastic bags, frozen and stored at -20°. All donor medical records were examined to ensure that there were no pre-existing pathologies, such as severe degenerative diseases or spinal pathologies, which could affect the structural responses of the specimens.

The muscular tissues were removed while keeping all the ligamentous structures intact. The lower cervical motion segments (C4-C5, C6-C7) were isolated, cleaned, and cast into aluminum cups with fiber-reinforced polyester resin. The cephalad end of the upper cervical spine specimens was secured using halo fixation. If necessary, the mandible and maxilla were removed in order to allow appropriate flexion range of motion. Upper cervical specimens (Head-C2) were inverted and mounted in the test frame using halo fixation of the head and casting of the C2 vertebra.

The test apparatus applies pure bending moments, and the resulting angular displacements are computed by tracking markers on digital images. The moments are generated using a pair of pneumatic pistons to apply a force-couple. After the specimen was mounted in the test frame, a counterweight was applied to the upper casting assembly. The mass of the counterweight was chosen to counterbalance the mass of the assembly, and to place the motion segment in 0.5 Newton of tension. The purpose of the 0.5 N tensile load was to establish an initial position for the motion segment in the middle of its neutral zone. The specimen was imaged in this position and the load cell values were zeroed. This image served as the reference position for the flexibility and the failure tests. Prior to flexibility testing, the specimens were preconditioned with 30 cycles of ±1.5 N-m of moment. The specimens were then loaded with pure flexion and extension moments in 0.5 N-m increments. Thirty seconds of creep were allowed prior to data acquisition, and the load was released between load steps. The peak applied moment was approximately 3.5 N-m. A Denton load cell is placed at the base of the specimen to collect forces and moments and to ensure that the moments remained pure. Lower range six-axis load cell (Denton model 2554A) was used for flexibility tests (FLEX and EXTN) and for failure of lower cervical segments C45 and C67. Higher range three-axis load cell (Denton model 1110AF-5K) was used for upper cervical spine (OC2) failure, which typically occurs at a higher moment than C45 and C67 failures.

After completion of the above bending tests, the tension mass was removed, and the specimens were failed in either flexion or extension. The loading rate was the maximum that could be achieved using our 100 psi system. The rates were dependent on the stiffness of the specimen, and were on the order of 90 N-m/second. The failures tests were imaged at 125 frames/second. Failure was defined as a decrease in the measured moment with increasing rotation, and they were verified by examination of the high speed images. Specimen dissection was performed to document injuries.

All the image data was processed with Matlab. This software enables us to digitize the location of the marker pins in each image. The location of each marker is determined using a centroid algorithm to increase accuracy. Angular displacements for each image are determined with respect to the reference image.